Joice Tang

UW HCDE Statement of Purpose

My research interests lie in Human-Computer Interaction (HCI), particularly in the areas of FATE (fairness, accountability, transparency, and ethics) and social computing pertaining to solidarity. As algorithms become more pervasive in our lives, there are many ways that technology can magnify inequities while also being used to resist their negative effects. Therefore, my interests have led me toward the following research directions:

1. How do **perceptions and folk theories of artificial intelligence (AI)** affect how people interact with AI systems, and how should we use these mental models to design systems thoughtfully?

Technology can be destructive even when creators have the best intentions. With complex AI systems, users often have unbalanced levels of trust in AI, either being overly trusting or skeptical of its advertised capabilities. This phenomenon can worsen existing societal inequities, especially in contexts such as education, social service distribution, criminal justice, and healthcare. Perhaps it can be argued that some of these systems should not be used or created, but the allure of efficiency and scale can lead to the development of these systems nonetheless. Therefore, I am interested in examining how people understand and interact with AI systems to influence more thoughtful system designs.

In Summer and Fall 2020, I joined Prof. Dongwook Yoon, Prof. Sidney Fels, and Dr. Kyoungwon Seo at UBC to study the perceived implications of AI's increasing presence in online classrooms. For this study, I worked closely with my co-authors to develop storyboards through extensive discussion and literature review before interviewing participants with the Speed Dating method. In addition to learning how to conduct robust qualitative research and analysis, my work on this project led to a co-first author publication in the International Journal of Educational Technology in Higher Education (Seo et al., 2021). In this paper, we discuss how scenarios were often ill-received if participants had misunderstandings about the capabilities of AI systems; this inspired me to explore how a lack of transparency in AI feedback could have tangible effects on users.

In Summer 2021, I began leading a project under the guidance of Prof. Edith Law at UWaterloo studying how people rationalize ambiguous or opaque AI feedback. Pursuing this research in the context of learning Chinese, I developed and piloted an experiment interface and survey/interview materials to probe how people react to opaque feedback about the aesthetics of their written Chinese. As I move toward future research, I hope to build upon the skills I have gained through these projects and continue to research how algorithmic transparency -- or lack thereof -- can affect stakeholders' understanding of complex systems.

2. How does technology change **power structures and agency in labor contexts**, and how might we design worker-centered systems?

In Fall 2020, I started working with Prof. Dongwook Yoon and Dr. Ning Ma to study worker-customer power asymmetry in food delivery platforms because of my interest in the ramifications of complex systems in sociotechnical contexts. In this project, I explored literature about gig work, algorithmic management, and worker-centered design, ultimately proposing a new research direction that led to a first-author 2021 paper at the IUI Workshop on Transparency and Explanations in Smart Systems (TExSS) (Tang et al., 2021). In this paper, we posit that the

lack of transparency in food delivery platforms might lead to an unfair attribution of accountability toward couriers in transaction breakdowns.

With the rise of gig work and side hustles, labor has become an even greater part of our everyday lives. At the same time, workplaces are becoming increasingly algorithmically managed and surveilled, and worker rights are declining. In addition to further alienating workers from their labor, algorithmic involvement can complicate power structures in workplaces, shifting responsibility and risk to workers and away from organizations. I am interested in researching how workers reclaim their agency through perspectives of collective action, policy, or worker-centered design.

3. How can technology be used toward **community organizing, community care, and collectively developing a critical understanding** of the world?

In my future research, I would like to explore how online interactions can help us support each other and understand how our concerns can be symptoms of systemic issues -- or, in the words of C. Wright Mills, to help us use our "sociological imagination." I am interested in analyzing how people currently work together to understand the world around them, and I would also like to design a system that constructively uses the power of community to foster understanding and solidarity. Specifically, I am interested in examining the role of technology in mutual aid contexts, in which communities work together to meet people's immediate needs while aiming to change the greater condition around them. In a world where algorithmic systems are becoming increasingly ubiquitous and influential, I hope to explore how online communities can collaborate to perform acts of resistance against an algorithmically-induced status quo. This can be a natural progression from my research regarding perceptions of AI and power asymmetries in labor, as it could easily be applied in the context of worker groups or cooperative resistance in digital environments (e.g. Irani & Silberman, 2013; Karizat et al., 2020).

UW HCDE and my future plans. At the University of Washington (UW), I am specifically interested in working with Professors Sucheta Ghoshal, Mark Zachry, and Benjamin Mako Hill. I admire their dedication to conducting thoughtful HCI research related to FATE and social change, and I can clearly see how my skills and interests fit in at UW. The interdisciplinary nature of the Human Centered Design and Engineering (HCDE) program would also allow me to learn from and collaborate with scholars with knowledge in a wide array of topics, which could help me become a stronger, more well-rounded researcher. Ultimately, I would like to become a professor or academic researcher who designs and analyzes technology through a lens of social responsibility, and I believe pursuing my Ph.D. in the HCDE program would significantly foster my development as a scholar in this direction.

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